MIL-C-48072 (MU)
2 June 1972
SUPERSEDING
PA-PD-3035
17 June 1968
(RENOVATION)

MILITARY SPECIFICATION

CAP, WINDSHIELD ASSEMBLY
METAL PARTS ASSEMBLY

1. SCOPE

1.1 Scope.-This Military Specification covers the renovation of the windshield cap (Drawing (dwg.) 9231607 is altered item dwg. to both cap windshield dwg. No's. 10534413 or 9209395), production of the nose plug and assembly of the nose plug to the windshield cap.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Fiberboard

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45208- Inspection Requirements, General Specification for
MIL-I-45607- Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1320

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MIL-STD-1167- Ammunition Data Card
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

ORDNANCE

9209395 - Cap, Windshield

10534413 - Cap, Windshield

9231607 - Cap, Windshield (Altered Item Drawing)

9231609 - Plug, Nose

9233448 - Cap, Windshield Assembly

9257510 - Cap, Windshield (Altered Item Drawing)

9257511 - Cap, Windshield Assembly

PUBLICATIONS

ORDNANCE CORPS

ET - 9257511 - Equipment Tabulation

3. REQUIREMENTS

3.1 General.-All parts shall meet the requirements of this document and the applicable drawings.

3.2 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of chips, cracks, dirt, grease, flux, rust, corrosion products and other foreign matter. The cleaning method shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Quality Conformance Inspection

4.2.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Nose plugs produced by one manufacturer in one unchanged process.

b. Windshield caps originally produced by one manufacturer from the same drawing, drawing revision, specification and specification revision, and renovated by one manufacturer in one unchanged process.

c. Assemblies produced in one unchanged process by one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying

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the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.3.

4.2.2.1 Plug, Nose (dwg. 9231609).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Threads damaged or not full	Visual	01001
102.	Thickness	Gage	01002
103.	Major diameter of threads, minimum (min.)	Gage	01003
Major:	AQL 0.40 percent		
104.	Pitch diameter of threads, min.	Gage	01004
Minor:	AQL 0.65 percent		
201.	Protective finish missing	Visual	01005

4.2.2.2 Cap, Windshield (Altered Item Dwg. 9257510, to be used in Conjunction with Windshield Cap Dwg. 10534413, Rev. E and Later).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Minor diameter of nose plug threads, maximum (max.)	Gage	02001
102.	Nose plug threads damaged or not full	Visual	02002

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Categories	Defects	Method of Inspection	Code No.
Major:	AQL 0.40 percent		
103.	Pitch diameter of nose plug threads, max.	Gage	02003
104.	True position of pitch diameter of nose plug threads to pitch diameter of windshield threads and rear face	Gage	02004
Minor:	None defined.		

4.2.2.3 Alternate Cap, Windshield (Altered Item Dwg. 9231607, to be used in Conjunction with Windhsield Cap Dwg. 10534413, Rev. D and Earlier, or Windshield Cap Dwg. 9209395).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Minor diameter of nose plug threads, max.	Gage	03001
102.	Nose plug threads damaged or not full	Visual	03002
Major:	AQL 0.65 percent		
103.	Pitch diameter of nose plug threads, max.	Gage	03003
104.	True position of nose plug thread pitch diameter	Gage	03004
105.	Chamfer missing	Visual	03005
Minor:	None defined.		

4.2.2.4 Cap, Windshield Assembly (Dwg. 9257511 or Alternate Dwg. 9233448) Prior to Assembly of Nose Plug.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Sealant not covering at least two full threads of nose plug threads	Visual	04001
Minor:	None defined.		

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4.2.2.5 Cap, Windshield Assembly (Dwg. 9257511 or Alternate Dwg. 9233448).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Stakes missing (two places) or not located in slot	Visual	05001
Major:	AQL 0.65 percent		
102.	Nose plug above flush	Gage	05002
103.	Nose plug exceeds below flush tolerance	Gage	05003
104.	Nose plug does not meet min. disassembly torque (See 4.2.4.1)	Test	05004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.2)	Visual	05005

4.2.3 Inspection Equipment.-Index of Inspection Equipment List Number ET-9257511 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.3.1 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.3.2 Supply and Maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

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4.2.4.3 Government Use of Contractor's Inspection and Test Equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.2.4 Test Methods and Procedures

4.2.4.1 Torque Test.-The disassembly torque shall be applied in a loosening direction while holding the parts immediately adjacent to the joints being tested so as to avoid possible loosening of any other joints. The joint under test shall be marked to provide a means for detection of any movement between the adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

5. PREPARATION FOR DELIVERY

5.1 Level C.-Packaging, packing and marking shall be in accordance with the requirements of MIL-STD-1169, and the following special instructions:

5.1.1 Packaging.-Two hundred (200) cap assemblies shall be packaged in a fiber box complete with plain pads and half-slotted partitions assembled to form individual cells. The cell arrangement shall be 40 per layer in an 8 x 5 arrangement - 5 layers per fiber box. A plain pad shall be placed on the bottom of the fiber box; between each layer of cells; and one or more pads on the top layer as required for a tight package. Cell size shall be 2 1/4 x 2 1/4 x 2 1/4 inch deep, inside dimensions. Box size shall be approximately 18 7/8 x 11 3/4 x 12 inches in depth, inside dimensions. Fiber boxes shall conform to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 200. Partitions and plain pads shall be of the same material as the box. Closure shall be as specified in the appendix of the box specification.

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6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Code Numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on the Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection Equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection Equipment Lists (EL).-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes, as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection Equipment List Codes.-The inspection equipment, as defined in 6.3.3 and 6.3.4, will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, Mandatory for Use

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6.3.3 Army Designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor Designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which completely depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of Contractor Designs.-All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of

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defects classified as Critical and Major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-A. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged; however, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1320-A137

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